Chapter 4
Magnetic Fields and Magnetic Forces

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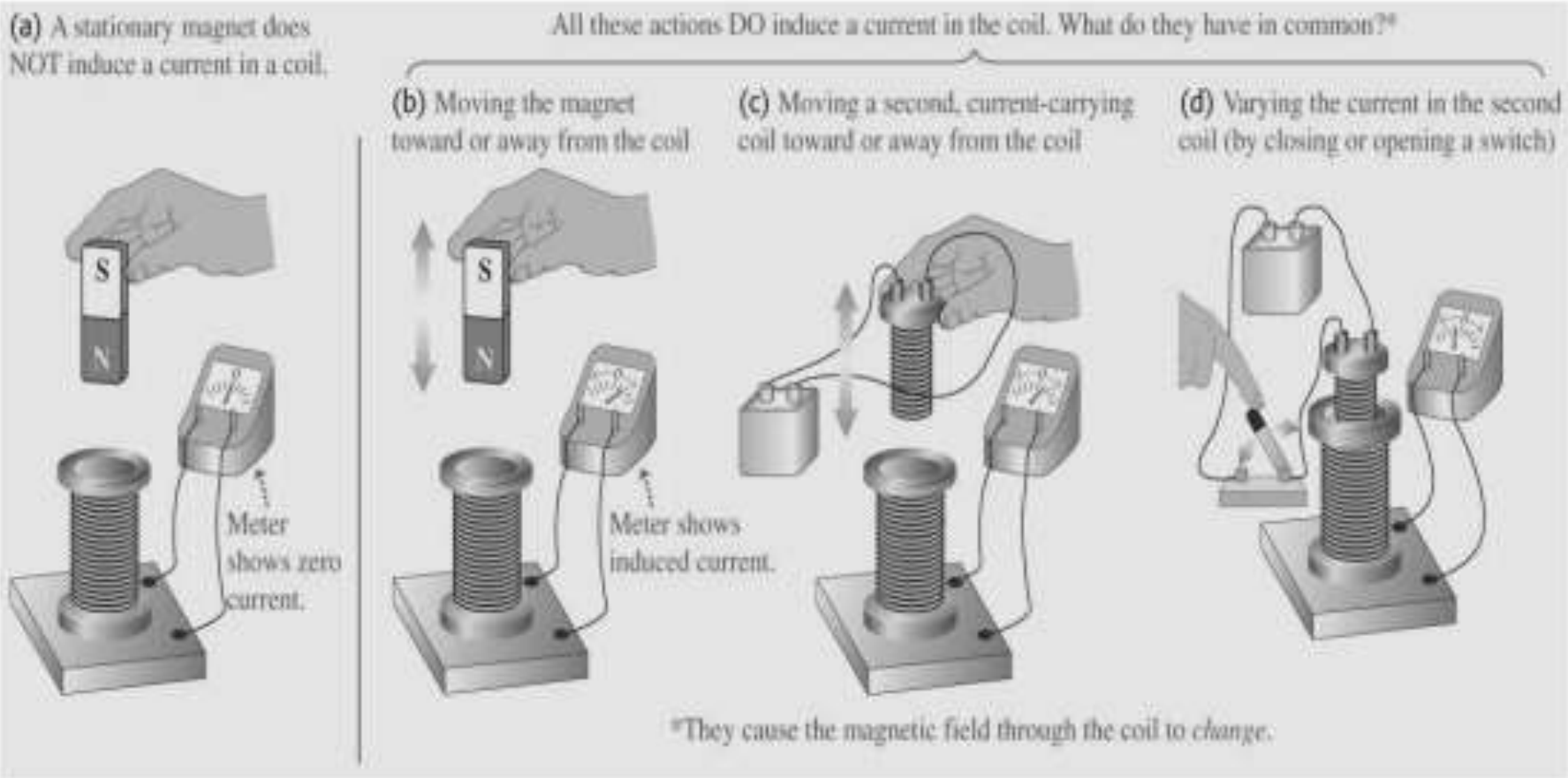
Electromagnetic Induction

1. Electromagnetic Experiments



This shows that the induced emf's that are causing the current do not depend on the material of the coil but only on its shape and the magnetic field.

Note:

The common element in all these experiments is changing magnetic flux through the coil connected to the galvanometer.

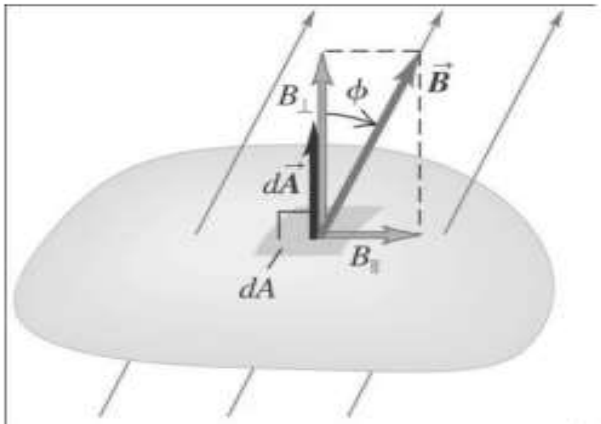
Electromagnetic Induction

Faraday's Law

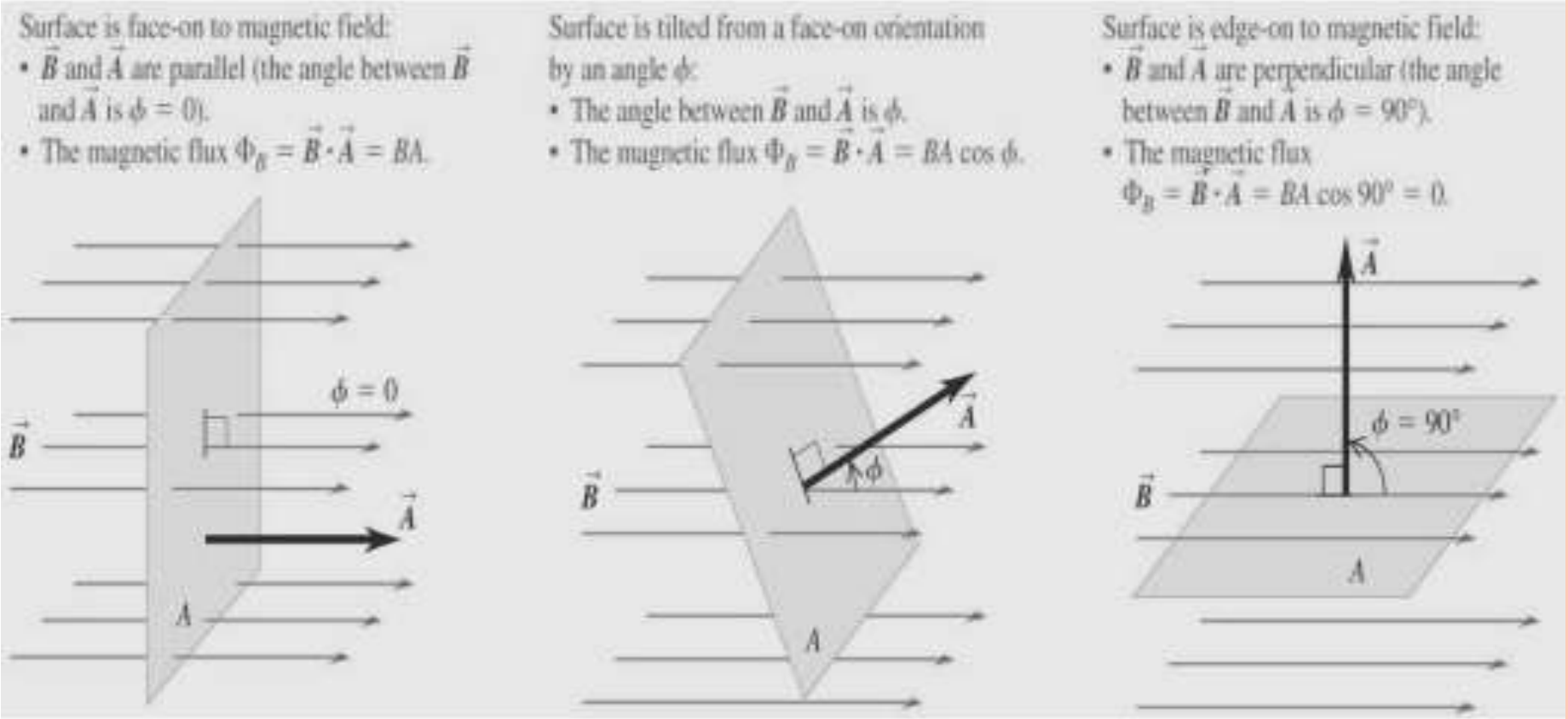
The common element in all induction effects is changing magnetic flux through the circuit. The magnetic flux is given by:

$$\Phi_B = \vec{B}.\vec{A} = BA\cos\varphi$$

If B is uniform over a flat area A .



Magnetic flux through element of area $d\vec{A}$:
$d\Phi_B = \vec{B} \cdot d\vec{A} = B_\perp dA = B\, dA \cos\phi$



Faraday's law of induction states that:

The induced emf in a closed loop equals the negative of the time rate of change of magnetic flux through the loop.

In symbols, Faraday's law is:

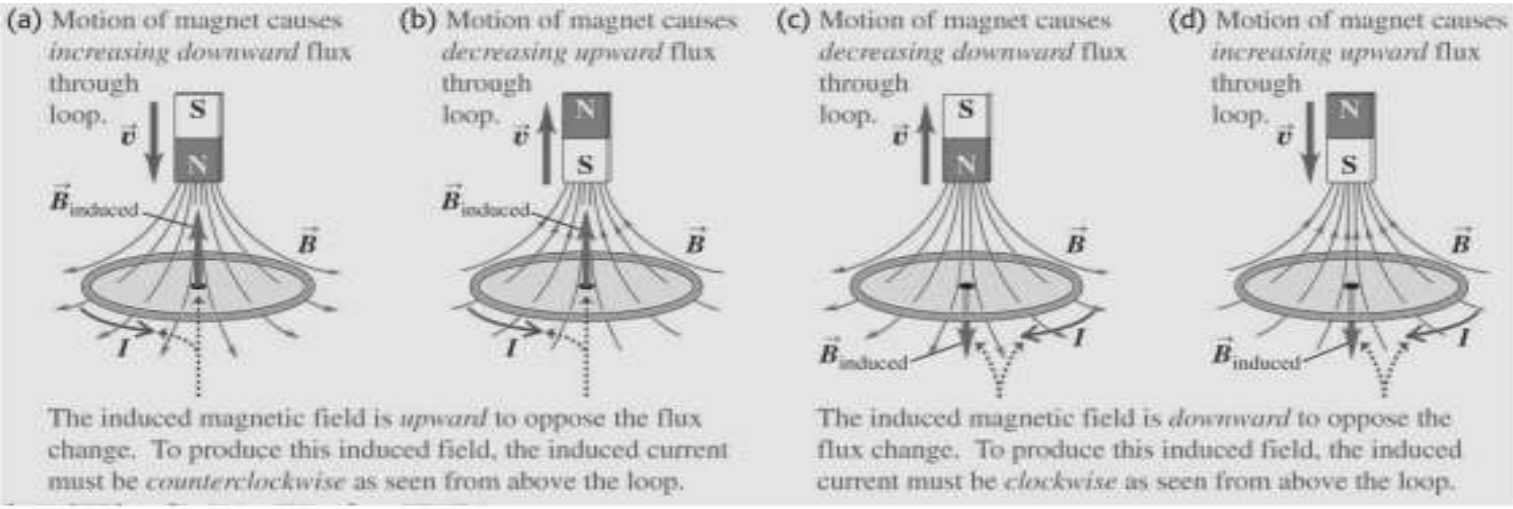
$$\varepsilon = -\frac{d\Phi_B}{dt}$$

If we have a coil with N identical turns, and if the flux varies at the same rate through each turn, the total rate of change through each turn is:

$$\varepsilon = -N\frac{d\Phi_B}{dt}$$

3. Lenz's Law

Lenz's Law is a convenient method for determining the direction of an induced current or emf. Lenz's law is not an independent principle: it can be derived from Faraday's law. Lenz's law states: The direction of any magnetic induction effect is such as to oppose the cause of the effect.



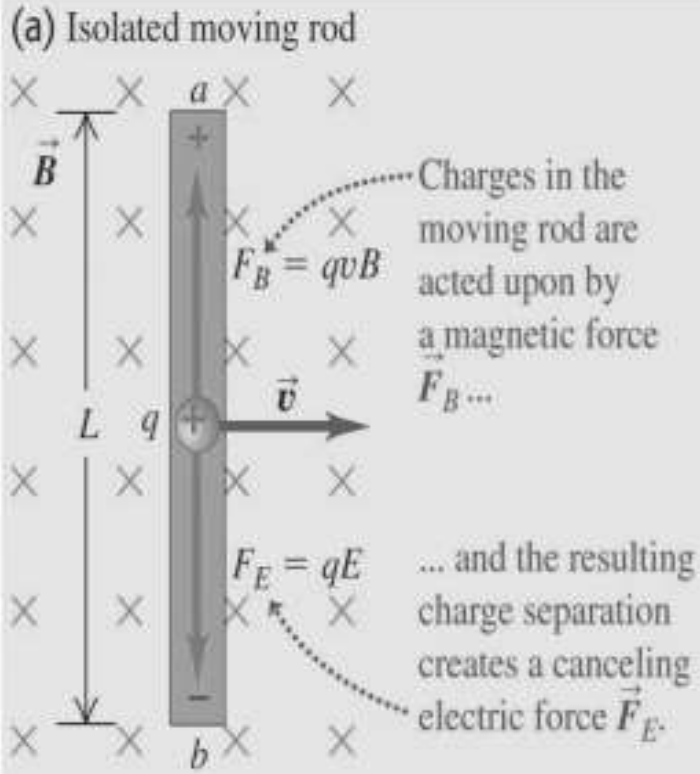
4. Motional Electromotive Force

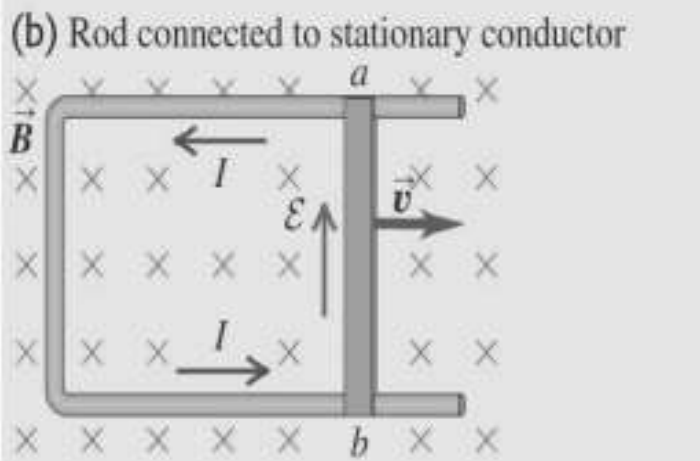
Figure shows the rod moves to the right at a constant velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ directed into the page. A charged particle q in the rod experiences a magnetic force with magnitude F =qvB. This magnetic force causes the free charges in the rod to move, creating a positive charge at the upper end and a negative charge at the lower end. This in turn creates an electric field $\vec{E}$ within the rod. There is a downward electric force with magnitude qE to cancel exactly the upward magnetic force with magnitude qvB. Then qE = qvB and the charges are in equilibrium. The magnitude of potential difference:

$$V_{ab} = V_a - V_b = EL = vBL$$

The moving rod has become a source of electromotive force. We call this emf a motional electromotive force, denoted by ε.

$$\varepsilon = vBL$$





The motional emf $\mathcal{E}$ in the moving rod creates an electric field in the stationary conductor.

MAGNETIC PROPERTIES OF MATERIALS

All matter is composed of atoms and atoms are composed of protons, neutrons and electrons. The protons and neutrons are located in the atom's nucleus and the electrons are in “constant motion” around the nucleus.

Electrons carry a negative electrical charge and produce a magnetic field as they move through space. A magnetic field is produced whenever an electrical charge is in motion. This may be hard to visualize on a subatomic scale but consider an electric current flowing through a conductor.

When electrons (electric current) are flowing through the conductor, a magnetic field forms around the conductor. The magnetic field can be detected using a compass. Since all matter is comprised of atoms, all materials are affected in some way by a magnetic field. However, not all materials react the same way.

At the atomic level, the motion of an electron gives rise to current loop
→ magnetic dipole moment → magnetic field

Magnetic dipole moment $\vec{m}$ [A.m^2]

$$\vec{m} = \vec{m}_u = N i A \vec{n}$$

Where N is the number of electrons, i is the current, and A is the area. It is a vector quantity and its direction can be found by the right hand screw rule.

The **magnetic moments** associated with atoms have three origins:
1 The electron orbital motion.
2 The change in orbital motion caused by an external magnetic field.
3 The spin of the electrons.

Magnetization $\vec{M}$ $[A.m^{-1}]$

It is equal to the magnetic dipole moment per unit volume

$$M = \frac{magnetic\ moment}{volume}$$

Permeability

Permeability is a material property that describes the ease with which a magnetic flux is established in the component. It is the ratio of the magnetic flux density to the magnetic intensity and, therefore, represented by the following equation: $\mu = \frac{B}{H}$

Magnetic susceptibility (χ_m)

It represents the response of a system to the external magnetic field. When a material is placed within a magnetic field, the material's electrons will be affected. However, materials can react quite differently to the presence of an external magnetic field.

This reaction is dependent on a number of factors such as the atomic and molecular structure of the material, and the net magnetic field associated with the atoms. In most atoms, electrons occur in pairs.

Each electron in a pair spins in the opposite direction, so when electrons are paired together, their opposite spins cause their magnetic fields to cancel each other.

Therefore, no net magnetic field exists. Alternately, materials with some unpaired electrons will have a net magnetic field and will react more to an external field.

Most materials can be classified as ferromagnetic, diamagnetic or paramagnetic.
Relation between the magnetic field (B), external magnetic field (H) and magnetization (M):

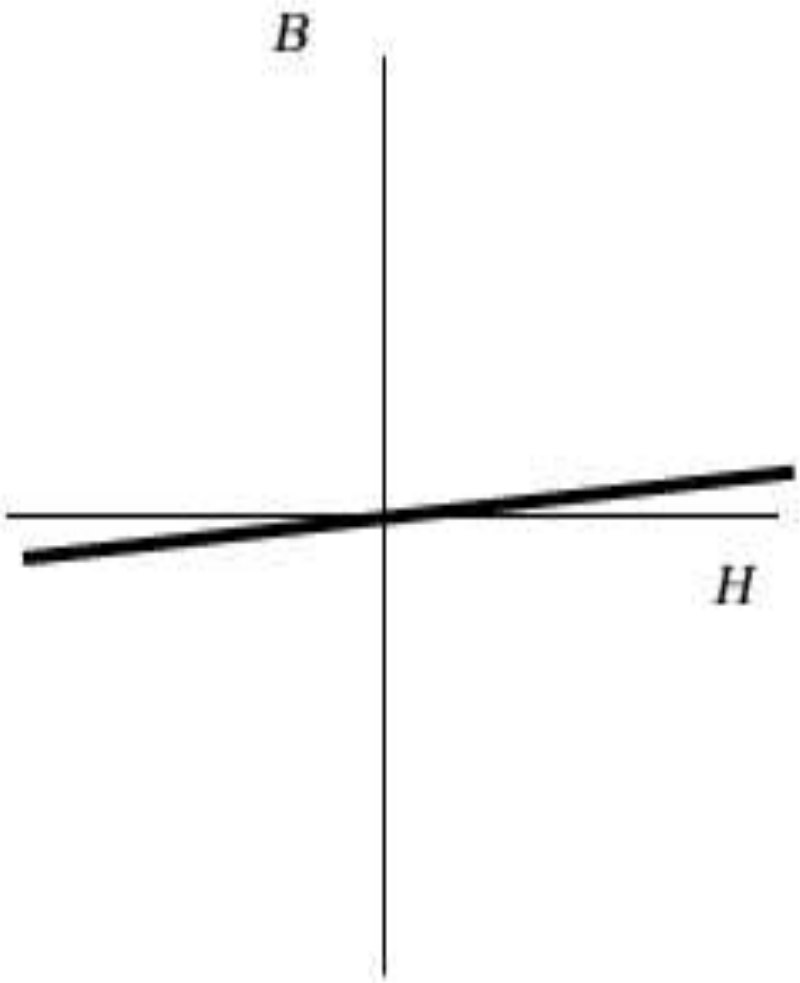
$$\vec{B} = \mu \vec{H} \qquad \vec{M} = \chi_m \vec{H} \qquad \mu = \mu_o (1 + \chi_m)$$

where
χ_m is the susceptibility, μ is the permeability, and μ_o is the permeability of free space (air).

<u>Diamagnetic materials</u> $\chi_m < 0$

- Small and negative susceptibility.
- Slightly repelled by a magnetic field.
- Do not retain the magnetic properties when the external field is removed.
- Magnetic moment – opposite direction to applied magnetic field.
- Solids with all electrons in pairs - no permanent magnetic moment per atom.

- Properties arise from the alignment of the electron orbits under the influence of an external magnetic field.
- Most elements in the periodic table, including copper, silver, and gold, are diamagnetic.
- χ_m(argon) ~ -1.0 x 10^{-8} χ_m (copper) ~ -1.0 x 10^{-5}



Diamagnetic material

$\chi_m < 0$ (small)

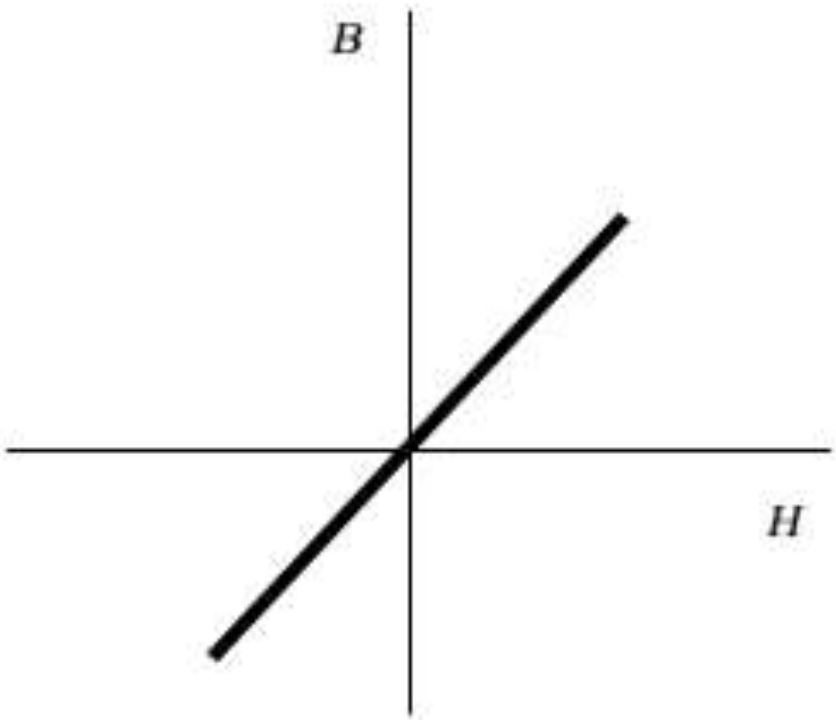
$$B = \mu_o (1 + \chi_m) H$$

Permeability

$\mu = \mu_o (1 + \chi_m)$ =slope of *B-H* line

Paramagnetic materials $\chi_m > 0$ small

- Small and positive susceptibility.
- Slightly attracted by a magnetic field.
- Material does not retain the magnetic properties when the external field is removed.
- Properties are due to the presence of some unpaired electrons and from the alignment of the electron orbits caused by the external magnetic field.
- Examples - magnesium, molybdenum, lithium, and tantalum.
- χ_m (oxygen) ~ 2.0×10^{-6} χ_m (aluminum) ~ 2.1×10^{-5}



Ideal magnetic material
or paramagnetic material
$\chi_m > 0$ (small)

$B = \mu_o \mu_r H = \mu H$

μ = constant = slope of B-H curve

Ferromagnetic materials

- Large and positive susceptibility.
- Strong attraction to magnetic fields.
- Retain their magnetic properties after the external field has been removed. Some unpaired electrons so their atoms have a net magnetic moment.
- Strong magnetic properties due to the presence of magnetic domains. In these domains, large numbers of atomic moments (10^{12} to 10^{15}) are aligned parallel so that the magnetic force within the domain is strong. When a ferromagnetic material is in the demagnetized state, the domains are nearly randomly organized and the net magnetic field for the part as a whole is zero. When a magnetizing force is applied, the domains become aligned to produce a strong magnetic field within the part.
- Iron, nickel, and cobalt are examples of ferromagnetic materials.
- $\vec{B} = \mu_o(\vec{H} + \vec{M})$ Magnetization is not proportional to the applied field.
- χ_m (ferrite) ~ 100 χ_m (iron) ~ 1000

Magnetic Domains

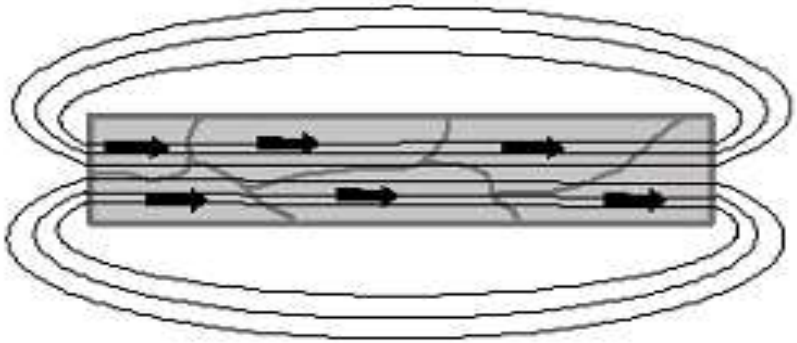
Ferromagnetic materials get their magnetic properties not only because their atoms carry a magnetic moment but also because the material is made up of small regions known as **magnetic domains**. In each domain, all of the atomic dipoles are coupled together in a preferential direction. Magnetic domains can be detected using Magnetic Force Microscopy (MFM).

Ferromagnetic materials become magnetized when the magnetic domains within the material are aligned. This can be done by placing the material in a strong external magnetic field or by passes electrical current through the material.

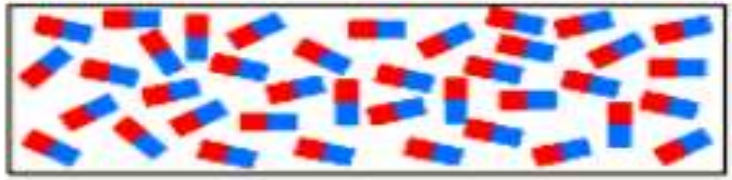
Some or all of the domains can become aligned. The more domains are aligned, the stronger the magnetic field in the material. When all of the domains are aligned, the material is said to be magnetically saturated.

When a material is magnetically saturated, no additional amount of external magnetization force will cause an increase in its internal level of magnetization.

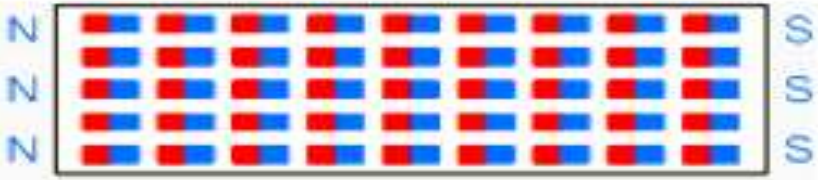
demagnetized Material



Magnetized Material



Loose and Individual
Magnetic Chain



Effect of Magnetization
Chains Lined-up in Series